

MVP2 UI/UX Design System Specification

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Revision 1 changelog: Added the mandatory OpenDesign cross-reference and token reconciliation (§1.2/§1.3); extended §7.2 with the full canonical 7-state connection lifecycle visual treatment (Reconnecting, KillSwitchActive, ConnectionFailed) plus a stateDiagram-v2 diagram; extended accessibility coverage with per-platform screen-reader bridge mechanics, dynamic type, and touch-target specs (§7.8–§7.11); added §11 Localization & i18n UX, §12 Biometric Auth Fallback UX, §13 Multi-Account/Profile Switching UX, §14 Enterprise SSO Login UX, and §15 Offline & Degraded-Network UI States (with a flowchart precedence diagram); extended Appendix B with per-phase accessibility gates and a new Phase 6.

Helix VPN Cross-Platform Design System

Version: 1.0.0

Date: July 2025

Status: Draft for Implementation

Scope: Desktop (macOS, Windows, Linux), Mobile (Android, iOS, HarmonyOS), Web, Browser Extension

Frameworks: Tauri v2 (Desktop), Flutter (Mobile), React (Web/Extension)

Table of Contents

1. [Design System Overview](#)

- 2. [Color System](#)
 - 3. [Typography](#)
 - 4. [Layout Grid](#)
 - 5. [Component Library](#)
 - 6. [Screen Specifications](#)
 - 7. [Interaction Patterns](#)
 - 8. [Iconography](#)
 - 9. [Animation & Motion](#)
 - 10. [Platform Adaptations](#)
 - 11. [Localization & Internationalization \(i18n\) UX](#)
 - 12. [Biometric Authentication Fallback UX](#)
 - 13. [Multi-Account / Profile Switching UX](#)
 - 14. [Enterprise SSO Login UX](#)
 - 15. [Offline & Degraded-Network UI States](#)
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1. Design System Overview

1.1 Design Philosophy

The Helix VPN design system follows a **clean, minimal, security-focused** philosophy. The UI prioritizes clarity and trust, reflecting the app's core purpose: protecting user privacy. Every design decision serves the goal of making security feel effortless and approachable.

Core Principles:

Principle	Description	Implementation
Clarity	Every element has a clear purpose	Minimal chrome, focused layouts, no decorative noise
Trust	Security should feel tangible	Color-coded states, clear visual feedback, transparent status
Efficiency	Primary actions are immediately accessible	One-tap connect, gesture shortcuts, contextual options
Consistency	Unified experience across all platforms	Shared design tokens, consistent interaction patterns
Accessibility	Usable by everyone	WCAG 2.1 AA compliance, screen reader support, keyboard navigation

1.2 Design Tokens Approach

Design tokens are the single source of truth for all visual properties. They enable cross-platform consistency while allowing platform-specific adaptations.

```
tokens/
├─ colors/                # Color palette tokens
│   ├── primary.json      # Brand colors (teal/cyan)
│   ├── semantic.json     # Status/state colors
│   ├── surfaces.json     # Background/surface colors
│   └─ text.json          # Text colors
├─ typography/           # Type scale tokens
│   ├── scale.json        # Size ramp (Display -> Overline)
│   └─ family.json        # Font family per platform
├─ spacing/              # Spacing scale tokens
│   └─ scale.json         # 4px base grid
├─ elevation/            # Shadow/depth tokens
│   └─ shadows.json
├─ motion/               # Animation tokens
│   ├── duration.json
│   └─ easing.json
└─ radii/                # Border radius tokens
    └─ scale.json
```

Token Format (JSON/CSS Variable):

```
/* Example: Primary token cascade */
--helix-primary-50: #E0F2F1;    /* Teal tint */
--helix-primary-100: #B2DFDB;
--helix-primary-200: #80CBC4;
--helix-primary-300: #4DB6AC;
--helix-primary-400: #26A69A;
--helix-primary-500: #00897B;   /* Primary brand */
--helix-primary-600: #00796B;
--helix-primary-700: #00695C;
--helix-primary-800: #004D40;
--helix-primary-900: #00332C;
```

OpenDesign is the mandatory token authority (not an inspirational reference). The tokens shown in this document are a platform-mapping restatement of the canonical, already-built **OpenDesign** system at [docs/design/README.md](#) — specifically [docs/design/opendesign/helix/DESIGN.md](#) (9-section spec), [docs/design/opendesign/helix/tokens.css](#) (canonical compiled CSS custom properties, --hx-* prefix), [docs/design/opendesign/helix/manifest.json](#), and [docs/design/opendesign/helix/components.html](#). Every one of the 8 platforms' UI implementations **MUST** build on these tokens — this is not one option among several. This document's role is narrower and downstream: it describes **how** the --hx-* OpenDesign tokens map onto each platform's native UI framework (React/Tauri consumes --hx-* CSS custom properties

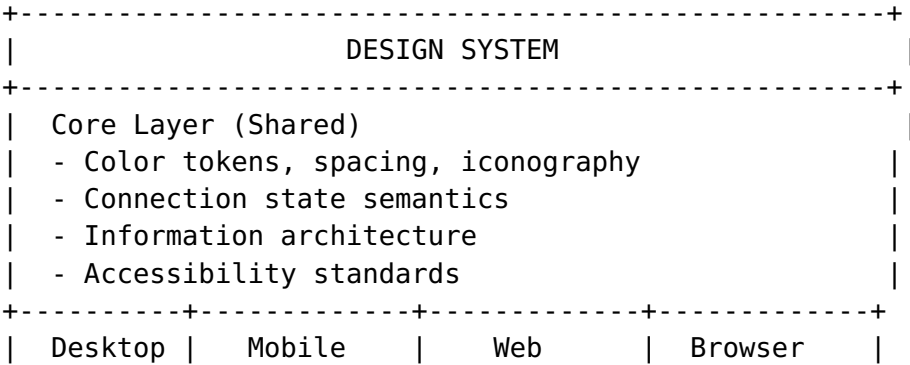
directly; Flutter/Dart's ThemeData/CupertinoThemeData are generated from the same token source; QML binds property declarations to them for Silica; the browser extension uses the same CSS variables) and the platform-specific interaction/screen-layout decisions built on top — it does **not** re-derive token values independently. Where this document uses an unprefixd or --helix-*-prefixed name (a naming convention that predates the OpenDesign consolidation, e.g. --helix-primary-500, --space-4, --radius-md), resolve it to the canonical --hx-* name in tokens.css (--helix-primary-500 → --hx-primary-500, --space-4 → --hx-space-4, --radius-md → --hx-radius-md) — values are identical, only the prefix differs, and tokens.css is authoritative on any future drift.

Token cross-check performed 2026-07-04: every color hex code (§2), typography size/weight/line-height/letter-spacing value (§3), and spacing/border-radius value (§4.3/§4.5) in this document was diffed against tokens.css and DESIGN.md. Result: colors, typography, spacing, and border-radius are **consistent** (byte-identical). Two discrepancies were found and reconciled: (1) Appendix A's motion quick-sheet listed Fast: 150ms against the canonical --hx-duration-fast: 100ms — corrected there, see the inline note at Appendix A; (2) §2.2's "Semantic Opacity Variants" listed only the light-theme 0.12 alpha, omitting tokens.css's distinct dark-theme 0.18 alpha override — noted inline at §2.2. No other mismatches found.

1.3 Cross-Platform Consistency Strategy

The design system employs a **"unified core with native skin"** approach:

- **Shared Design Tokens:** Colors, spacing, typography scales are consistent across all platforms
- **Platform-Native Components:** Widgets adapt to each platform's design language (Material 3, Cupertino, Fluent, Silica)
- **Unified Interaction Patterns:** Connection flows, state transitions, and gestures are consistent
- **Platform-Specific Chrome:** Navigation, menus, and system integration follow platform conventions



(Tauri)	(Flutter)	(React)	Extension
- Web UI	- Material3	- Tailwind	- Popup UI
- System	- Cupertino	+ shadcn	- Options
tray	- ArkUI		- Content
- Menu	- Silica		script
+-----+	+-----+	+-----+	+-----+

OpenDesign is the Core Layer. The "Core Layer (Shared)" box above — color tokens, spacing, iconography, connection-state semantics, accessibility standards — **is** the OpenDesign token system (docs/design/README.md, §1.2 above), not a parallel or duplicate abstraction maintained independently in this document. Every platform-specific "Design Adjustment" in §10 is layered strictly on top of unmodified OpenDesign tokens: a platform adaptation may change layout, chrome, density, or interaction convention, but **MUST NOT** redefine a color, spacing, typography, or motion token value — e.g. Windows's 8px spacing density (§10.2) or macOS's 16px card radius (§10.1) are *component-level* choices built from the same `--hx-space-*/--hx-radius-*` primitives, never a fork of the primitive itself.

1.4 Accessibility First (WCAG 2.1 AA)

All components and screens must meet **WCAG 2.1 Level AA** compliance:

Requirements:

- Color contrast ratio $\geq 4.5:1$ for body text
 - Color contrast ratio $\geq 3:1$ for large text (18pt+) and UI components
 - All interactive elements have visible focus indicators (focus-visible)
 - All form inputs have associated `<label>` elements
 - Dynamic content changes announced via `aria-live` regions
 - Full keyboard navigation support (Tab order, Enter/Space activation)
 - Touch targets minimum 44x44dp/dpx (mobile)
 - `prefers-reduced-motion` respected for all animations
 - Screen reader tested with NVDA (Windows), JAWS (Windows), VoiceOver (macOS/iOS), TalkBack (Android)
-

2. Color System

2.1 Primary Palette (Teal/Cyan)

The primary palette uses teal/cyan tones to evoke trust, security, and technological sophistication.

Token	Hex	HSL	Usage
--helix-primary-50	#E0F2F1	174° 52% 91%	Lightest tint, backgrounds
--helix-primary-100	#B2DFDB	174° 47% 78%	Hover states, light fills
--helix-primary-200	#80CBC4	174° 42% 65%	Secondary highlights
--helix-primary-300	#4DB6AC	174° 38% 51%	Accent elements
--helix-primary-400	#26A69A	174° 62% 40%	Links, active states
--helix-primary-500	#00897B	174° 100% 27%	Primary brand color
--helix-primary-600	#00796B	174° 100% 24%	Primary hover
--helix-primary-700	#00695C	174° 100% 21%	Primary pressed
--helix-primary-800	#004D40	174° 100% 15%	Strong emphasis
--helix-primary-900	#00332C	174° 100% 10%	Deepest shade

Secondary Cyan Accent:

Token	Hex	Usage
--helix-accent-400	#00BCD4	Secondary accent, data viz
--helix-accent-500	#00ACC1	Accent hover
--helix-accent-600	#0097A7	Accent pressed

2.2 Semantic Colors

Semantic colors convey meaning universally across all platforms.

State	Token	Hex	RGBA	Usage
Connected	--helix-connected	#4CAF50	rgba(76,175,80,1)	VPN connected,

State	Token	Hex	RGBA	Usage
				success states
Connecting	--helix-connecting	#FF9800	rgba(255,152,0,1)	In-progress, warning states
Disconnected	--helix-disconnected	#F44336	rgba(244,67,54,1)	VPN disconnected, offline
Error	--helix-error	#D32F2F	rgba(211,47,47,1)	Critical errors, destructive
Warning	--helix-warning	#F57C00	rgba(245,124,0,1)	Caution, attention needed
Info	--helix-info	#1976D2	rgba(25,118,210,1)	Informational messages

Semantic Opacity Variants:

```
--helix-connected-bg: rgba(76, 175, 80, 0.12);
--helix-connecting-bg: rgba(255, 152, 0, 0.12);
--helix-disconnected-bg: rgba(244, 67, 54, 0.12);
--helix-error-bg: rgba(211, 47, 47, 0.12);
--helix-warning-bg: rgba(245, 124, 0, 0.12);
--helix-info-bg: rgba(25, 118, 210, 0.12);
```

Cross-checked against OpenDesign 2026-07-04 (§1.2): these are the light-theme (default) alpha values and are consistent with tokens.css. The dark theme applies a distinct, intentionally higher alpha (**0.18**, not a re-inversion of 0.12) for these same six semantic backgrounds — see tokens.css's [data-theme="dark"] block — for adequate visibility against dark surfaces. Implementers **MUST** reference --hx-semantic-*-bg (theme-aware) rather than hardcoding 0.12 in a dark-theme code path.

2.3 Latency Color Coding

Latency indicators use a traffic-light system:

Latency Range	Color	Hex	Semantic
< 50ms	Green	#4CAF50	Excellent
50-100ms	Amber	#FF9800	Good
100-200ms	Orange	#F57C00	Fair
> 200ms	Red	#F44336	Poor

2.4 Dark Theme Colors

```
.dark {  
  /* Surfaces */  
  --helix-bg-primary: #0A1929;      /* Deepest background */  
  --helix-bg-secondary: #132F4C;    /* Cards, panels */  
  --helix-bg-tertiary: #1E4976;     /* Elevated surfaces */  
  --helix-bg-elevated: #243B55;     /* Modals, dialogs */  
  
  /* Text */  
  --helix-text-primary: #F0F4F8;    /* Primary text */  
  --helix-text-secondary: #8BA3B8;  /* Secondary/muted */  
  --helix-text-tertiary: #5A7A94;   /* Placeholder, disabled */  
  --helix-text-disabled: #3D5A80;   /* Disabled text */  
  
  /* Borders */  
  --helix-border-default: #1E4976;  
  --helix-border-subtle: #132F4C;  
  --helix-border-focus: #00897B;  
}
```

2.5 Light Theme Colors

```
.light {  
  /* Surfaces */  
  --helix-bg-primary: #F0F4F8;      /* Background */  
  --helix-bg-secondary: #FFFFFF;    /* Cards, panels */  
  --helix-bg-tertiary: #E8EDF2;     /* Elevated surfaces */  
  --helix-bg-elevated: #FFFFFF;     /* Modals, dialogs */  
  
  /* Text */  
  --helix-text-primary: #0A1929;    /* Primary text */  
  --helix-text-secondary: #4A6375;  /* Secondary/muted */  
  --helix-text-tertiary: #8BA3B8;   /* Placeholder */  
  --helix-text-disabled: #B0C4D4;   /* Disabled text */  
  
  /* Borders */  
  --helix-border-default: #C8D8E4;  
  --helix-border-subtle: #E2EBF2;  
  --helix-border-focus: #00897B;  
}
```

2.6 Surface & Background Colors Summary

Surface	Dark Theme	Light Theme
App background	#0A1929	#F0F4F8
Card/panel	#132F4C	#FFFFFF
	#1E4976	#E8EDF2

Surface	Dark Theme	Light Theme
Elevated (modal)		
Input field	#0D2137	#FFFFFF
Selected item	rgba(0,137,123,0.15)	rgba(0,137,123,0.10)
Hover overlay	rgba(255,255,255,0.05)	rgba(0,0,0,0.04)
Pressed overlay	rgba(255,255,255,0.08)	rgba(0,0,0,0.08)
Divider	#1E4976	#E2EBF2
Scrim (modal backdrop)	rgba(0,0,0,0.7)	rgba(0,0,0,0.5)

3. Typography

3.1 Font Family Selection

Platform	Font Stack	Fallback
Desktop (Tauri)	system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif	sans-serif
Android (Flutter)	Roboto (Material 3 default)	Device sans-serif
iOS (Flutter)	SF Pro Text/Display (Cupertino)	Device sans-serif
HarmonyOS	HarmonyOS Sans	Device default
Aurora OS	Sailfish Silica	System font
Web/ Extension	Inter, system-ui, -apple-system, sans-serif	sans-serif

Monospace (Technical Data): | Platform | Font | Usage |
|---|---|---| | All | JetBrains Mono, 'Fira Code', 'SF Mono', monospace |
IP addresses, connection stats, protocol info, keys |

3.2 Type Scale

The type scale follows a 1.25x ratio (major third) with a 14px base.

Token	Size	Weight	Line Height	Letter Spacing	Usage
Display			1.2	-0.5px	

Token	Size	Weight	Line Height	Letter Spacing	Usage
Headline	36px / 2.25rem	300 (Light)	1.3	-0.25px	Connection status numbers
	28px / 1.75rem	600 (SemiBold)			Screen titles (desktop)
	22px / 1.375rem	600			Dialog titles, section headers
Title Medium	18px / 1.125rem	600	1.4	0.15px	Card titles, sub-screens
Title Small	16px / 1rem	500 (Medium)	1.4	0.1px	List section headers
Body Large	16px / 1rem	400 (Regular)	1.5	0.5px	Primary body text
Body Medium	14px / 0.875rem	400	1.5	0.25px	Default body, descriptions
Body Small	12px / 0.75rem	400	1.5	0.4px	Captions, metadata
Caption	11px / 0.6875rem	500	1.4	0.5px	Labels, timestamps
Overline	10px / 0.625rem	600	1.4	1.5px	Category labels, all-caps

3.3 Font Weights

Weight	Name	Usage
300	Light	Large display numbers, hero text
400	Regular	Body text, descriptions
500	Medium	Buttons, list items, emphasis
600	SemiBold	Headlines, titles, section headers
700	Bold	Critical labels, connection status

3.4 Monospace Type Scale (Technical Data)

Token	Size	Weight	Usage
Mono Large	18px	400	Connection time, large stats
Mono Medium	14px	400	IP addresses, server info

Token	Size	Weight	Usage
Mono Small	12px	500	Protocol badges, key fragments
Mono Caption	11px	400	Debug info, logs

4. Layout Grid

4.1 Desktop Layout Grid

The desktop UI uses a responsive grid system with a minimum width of 360px and maximum of 1440px.

Property	Value
Min width	360px (small popups)
Max width	1440px (full admin panel)
Main window	420px (connection) / 520px (servers) / 640px (settings)
Gutter	24px
Column count	12 (fluid)
Margin	24px (desktop), 16px (compact)

Desktop Window Sizes: | Screen | Width | Height | Purpose |
|---|---|---|---| | Main (connected) | 420px | 600px | Primary interface
| | Main (disconnected) | 420px | 480px | Compact state | | Server
Selection | 520px | 700px | Full server list | | Settings | 640px |
700px | Tabbed settings | | Connection Details | 420px | 550px |
Stats overlay |

4.2 Mobile Layout Grid

Platform	System	Base Unit
Android	Material 3	8dp grid
iOS	Cupertino	8pt grid
HarmonyOS	ArkUI	4vp grid

Mobile Safe Areas:

- Top: Respect SafeArea top inset (notch, status bar)
- Bottom: Respect SafeArea bottom inset (home indicator, nav bar)
- Horizontal: 16dp/dpx padding (mobile), 24dp (tablet)

4.3 Spacing Scale (4px Base)

All spacing uses a 4px base unit for consistency.

Token	Value	Pixels	Usage
--space-0-5	0.125rem	2px	Icon gaps, hairline spacing
--space-1	0.25rem	4px	Tight component internal padding
--space-2	0.5rem	8px	Default component internal padding
--space-3	0.75rem	12px	Small component margin
--space-4	1rem	16px	Default padding, card gutters
--space-5	1.25rem	20px	Medium spacing
--space-6	1.5rem	24px	Section padding, dialog margins
--space-8	2rem	32px	Large section gaps
--space-10	2.5rem	40px	Section separators
--space-12	3rem	48px	Major section padding
--space-16	4rem	64px	Hero spacing, connection button margin
--space-20	5rem	80px	Large visual separation

4.4 Responsive Breakpoints

Token	Width	Target
xs	< 380px	Extension popup (minimum)
sm	380-640px	Extension popup, mobile PWA
md	640-768px	Tablet PWA, small panels
lg	768-1024px	Tablet landscape, compact desktop
xl	1024-1280px	Desktop standard
2xl	1280px+	Desktop wide, admin panel

4.5 Border Radius Scale

Token	Value	Usage
--radius-none	0	Sharp edges (tables, dividers)
--radius-sm	4px	Small tags, badges
--radius-md	8px	Buttons, inputs, small cards
--radius-lg	12px	Cards, panels
--radius-xl	16px	Large cards, bottom sheets

Token	Value	Usage
--radius-2xl	24px	Modals, dialogs
--radius-full	9999px	Pills, avatars, connection button

5. Component Library

5.1 Desktop Components (Tauri/WebView)

5.1.1 Button Variants

Variant	Background	Text	Border	Hover
Primary	#00897B	#FFFFFF	none	#007
Secondary	transparent	#00897B	1px #00897B	rgba(
Ghost	transparent	#F0F4F8	none	rgba(
Danger	#D32F2F	#FFFFFF	none	#B71
Disabled	rgba(255,255,255,0.12)	rgba(255,255,255,0.38)	none	-

Button Specs:

- Min height: 40px (standard), 32px (compact)
- Horizontal padding: 24px (standard), 16px (compact)
- Border radius: 8px (standard), 4px (compact)
- Font: Body Medium, 500 weight
- Icon + text gap: 8px

5.1.2 Connection Toggle (Desktop)

+-----+
[Power Icon]
CONNECTED
US East (23ms)
+-----+

- Size: 140px x 140px circular
- Connected: Fill #00897B with #4CAF50 glow pulse
- Connecting: Animated spinner on #FF9800 ring
- Disconnected: #F44336 with subtle shadow
- Border: 4px solid ring (color = state)
- Glow: 20px blur, state color at 30% opacity, pulsing
- Icon: 48px, white
- Label below: Body Large, 600 weight, state color

5.1.3 Server List Item

+-----+									
	[FLAG]	US	East	-	New York		[Fav]	WG	
		23ms	*	*	*	*	34% load		
+-----+									

- Height: 64px
- Left: Flag icon (36x36px, 8px radius) + 12px gap
- Title: Body Large, 500 weight
- Subtitle: Body Small, secondary color
- Right: Favorite star + Protocol badge + Chevron
- Divider: 1px bottom border
- Selected state: Primary tint background (10% opacity)
- Hover: Hover overlay

5.1.4 Protocol Badge

+-----+		
	WG	
+-----+		

- Size: auto-width x 20px
- Background: Primary at 10% opacity
- Text: Primary color, Caption weight 500
- Border radius: 4px
- Padding: 0 6px
- Variants: "WG" (WireGuard), "OV" (OpenVPN), "IK" (IKEv2)

5.1.5 Latency Indicator

Color	Size	Animation
Green (<50ms)	8px circle	Static
Amber (50-100ms)	8px circle	Subtle pulse
Orange (100-200ms)	8px circle	Pulse
Red (>200ms)	8px circle	Fast pulse

- Displayed as colored dot + "23ms" label
- Dot has 4px colored glow (same color, 40% opacity)

5.1.6 Settings Card/Section

- Background: bg-secondary
- Border radius: 12px
- Padding: 16px
- Section header: Title Small, primary color, uppercase
- Item height: 56px
- Divider between items: 1px border-subtle

- Left icon: 24px, primary or secondary color

5.1.7 Input Fields

```
+-----+
| Label                                     |
| +-----+                               |
| | Placeholder text           [] |      |
| +-----+                               |
| Helper text                             |
+-----+
```

- Height: 48px
- Background: bg-primary (dark) / #FFFFFF (light)
- Border: 1px border-default
- Border radius: 8px
- Padding: 0 16px
- Font: Body Medium
- Focus: 2px border-focus outline
- Error: 2px #D32F2F outline + red helper text
- Disabled: 38% opacity

5.1.8 Modal/Dialog

- Min width: 360px, Max: 560px
- Background: bg-elevated
- Border radius: 16px
- Shadow: 0 24px 48px rgba(0,0,0,0.4)
- Header: Title Large, 600 weight
- Content: Body Medium
- Actions: Right-aligned, Primary + Secondary buttons
- Backdrop: Scrim color with fade-in 150ms

5.1.9 Toast Notification

```
+-----+
| [Icon] Message                               [X] |
+-----+
```

- Width: auto (max 400px)
- Height: 48px
- Background: bg-elevated
- Border left: 4px (semantic color)
- Border radius: 8px
- Shadow: 0 8px 24px rgba(0,0,0,0.3)
- Duration: 4000ms (auto-dismiss)
- Animation: Slide in from top-right, 300ms ease-out

5.1.10 System Tray Menu (Desktop)

+-----+		
	HelixVPN - Connected	
	US East (23ms)	

	[ICON] Connect	
	[ICON] Disconnect	

	[ICON] Select Server...	
	[ICON] Preferences...	

	[ICON] Quit	
+-----+		

- Follows platform menu conventions
 - macOS: NSStatusItem with template icon
 - Windows: NotifyIcon with context menu
 - Linux: AppIndicator/StatusNotifierItem
 - Items: 28px height, 16px icon, 12px gap
 - Separator: 1px line
-

5.2 Mobile Components (Flutter)

5.2.1 Connection Button (Large Circular)

- Size: 140dp diameter
- Shape: Circle with radial gradient glow
- Connected: #00897B fill, #4CAF50 pulsing outer ring (20dp spread)
- Connecting: #FF9800 fill, rotating spinner overlay
- Disconnected: #F44336 fill, static
- Icon: 48dp, white, centered
- Shadow: 0 20px 40px state-color at 30%
- Animation: Pulse scale 1.0 -> 1.08, 1500ms ease-in-out (connected only)

5.2.2 Server Selection Tile

- Height: 72dp (Material 3 ListTile)
- Leading: Country flag circle (40dp)
- Title: Body Large, 500 weight
- Subtitle: Body Small (latency dot + "23ms" + load %)
- Trailing: Protocol chips + favorite icon
- Selected: Primary tinted background
- Ripple: Platform default

5.2.3 Bottom Sheet

- Initial height: 50% screen
- Max height: 90% screen
- Background: Surface color
- Top handle: 4dp x 32dp, #8BA3B8, 2dp radius
- Border radius: 24dp top corners
- Entry animation: Slide up 300ms, ease-out
- Drag to dismiss: Velocity threshold 300dp/s

5.2.4 Tab Bar

- Height: 48dp (Android) / Cupertino navigation bar (iOS)
- Background: Surface color
- Indicator: 2dp height, Primary color
- Active text: Primary, 500 weight
- Inactive text: Secondary color, 400 weight
- iOS: Segmented control or bottom Cupertino tab bar

5.2.5 Switch/Toggle (Platform-Adaptive)

- Android: Material 3 Switch with Primary track
- iOS: CupertinoSwitch with Primary active color
- Track: 32dp x 52dp (iOS) / 28dp x 48dp (Android)
- Thumb: 24dp (iOS) / 20dp (Android)
- Active: Primary color track
- Inactive: Surface variant track

5.2.6 Text Field (Adaptive)

- Android: Material 3 OutlinedTextField with Primary focus
- iOS: CupertinoTextField with subtle border
- Height: 56dp (Android) / 44dp (iOS)
- Border radius: 8dp
- Focus: 2dp Primary outline
- Error: 2px #D32F2F outline

5.2.7 Card

- Background: Surface color
- Border radius: 16dp
- Elevation: 0 (flat), shadow for elevated
- Padding: 16dp
- Content: Title + body + optional actions

5.2.8 Speed Graph/Sparkline

- Height: 80dp
- Width: Parent full width
- Download line: #00897B, 2dp stroke

- Upload line: #00BCD4, 2dp stroke, dashed
- Fill: Gradient fade to transparent (line color 20%)
- Grid: Dotted horizontal lines at 25/50/75%
- Labels: Mono Caption, secondary color
- Update interval: 1000ms

5.2.9 Data Usage Indicator

+-----+		
Download	Upload	
[==== 1.2 GB]	[==== 340 MB]	
45% of limit		
+-----+		

- Two-column layout
- Value: Mono Medium, 500 weight
- Bar: 4dp height, full border radius
- Fill: Primary (download), Accent (upload)
- Background: Surface variant

5.2.10 Quick Settings Tile (Android)

- Size: 1x1 tile (standard QS tile)
- Active: Primary color background, white icon
- Inactive: Surface color, secondary icon
- Label: "Helix VPN" / "Helix VPN: ON"
- Toggle animation: Color crossfade 200ms
- Long-press: Opens app

5.3 Aurora OS Components (Silica)

5.3.1 Pulley Menu

- Trigger: Pull down from top edge
- Background: Primary color, dark theme
- Items: 56px height, white text
- Icon: 24px left-aligned
- Selection highlight: rgba(255,255,255,0.15)
- Animation: Slide down from top, items stagger 50ms

5.3.2 Context Menu

- Trigger: Long-press on list items
- Background: Elevated surface
- Items: 48px height, Body Medium
- Divider: 1px subtle
- Animation: Fade in 150ms + scale from anchor point

5.3.3 List Item (Silica Style)

- Height: 80px (standard), 64px (compact)
- Background: Transparent
- Primary label: Body Large, text-primary
- Secondary label: Body Small, text-secondary
- Icon: 32px, left-aligned, 16px padding
- Divider: 1px bottom border-subtle
- Pressed: rgba(255,255,255,0.08) overlay

5.3.4 Text Field (Silica Style)

- Background: Surface color
- Border: 1px border-default, no radius (Silica rectangular)
- Height: 48px
- Font: Body Medium
- Placeholder: text-tertiary
- Focus: 2px Primary underline
- Error: 2px #D32F2F underline

5.3.5 Slider

- Track height: 4px
- Active track: Primary color
- Inactive track: Surface variant
- Thumb: 20px circle, Primary fill, 2px white border
- Value label: Floating tooltip above thumb

5.3.6 Dialog (Silica)

- Width: 90% screen, max 480px
- Background: Elevated surface
- Border radius: 8px (Silica style)
- Header: Title Medium, 600 weight
- Content: Body Medium
- Actions: Full-width buttons stacked
- Entry: Fade 200ms + slight scale

5.3.7 Cover Page

- Size: 1:1 square (app cover)
 - Background: Gradient from Primary-800 to Primary-900
 - Connection status: Centered large dot (12px) + state text
 - Server: Caption text below status
 - Animation: Dot pulse glow matching state color
 - Ambiance: Respects system ambiance settings
-

6. Screen Specifications

6.1 Desktop Screens

6.1.1 Main Window (Connected State)

```
+-----+
| [TRAY] Helix VPN [ ] [X] |
+-----+
|
| [Server Selector Bar]
| [US Flag] US East - New York >
|
|           +-----+
|           | [POWER] | <- Large
|           | CONNECTED| toggle
|           +-----+
|           (glowing ring)
|
| Connected
| 23ms latency
|
| IP: 203.0.113.45
| Protocol: WireGuard
| Duration: 02:34:18
|
| [Speed graph sparkline]
|
| Download: 1.2 GB Upload: 340MB
|
+-----+
```

Specs:

- Window: 420px x 600px
- Background: bg-primary
- Server selector: 64px height, bg-secondary, 12px radius
- Connection button: 140px circle, centered
- Status text: Title Medium, #4CAF50
- Info panel: bg-secondary card, 12px radius, Mono Small text
- Speed graph: 80px height, full width minus 32px padding
- Data usage: Two-column flex layout

6.1.2 Main Window (Disconnected State)

- Window: 420px x 480px (compact)
- Connection button: 140px, #F44336 fill
- Status: "Disconnected", Title Medium, #F44336
- Info panel hidden

- Prompt: "Tap to connect securely", Body Small, secondary

6.1.3 Server Selection

- Window: 520px x 700px
- Search bar: Sticky top, 48px height
- Quick connect: "Optimal Location" tile, 56px
- Section headers: "Favorites", "All Locations" - Overline style
- Server list: Scrollable, items 64px height
- Country groups: Collapsible accordion
- Latency sort: Default (ascending)
- Selected: Primary tint background + checkmark

6.1.4 Settings (Tabbed)

- Window: 640px x 700px
- Tabs: General | Connection | Account | Advanced
- Tab height: 40px
- Content: Settings cards in vertical scroll
- Each card: bg-secondary, 12px radius, 16px padding
- Card title: Title Small, primary color
- Card items: 56px height rows

6.1.5 Connection Details / Stats

- Overlay modal or separate tab
- Real-time: Download/upload speed (Display size)
- Connection time: Mono Large, counting up
- Protocol info: Badge + details
- Server info: Flag + name + IP
- Encryption: Cipher info in monospace
- Graphs: 5-minute rolling sparkline

6.2 Mobile Screens

6.2.1 Connection (Home Screen)

```
+-----+
| [Menu] Helix VPN  [Cog]  |
+-----+
|
|  [Server Bar]
|  [US] US East          23ms |
|
|          +-----+
|          |  0  |
|          +-----+
|      CONNECTED
|
```

[Connection Stats Card]
IP: 203.0.113.45
Time: 02:34:18
Protocol: WireGuard
[Speed Sparkline]
DL: 45 Mbps UL: 12 Mbps

+-----+

Specs:

- Full screen, SafeArea respected
- App bar: 56dp, transparent or surface
- Server bar: 72dp, tappable, navigates to server list
- Connection button: 140dp circle, centered vertically
- Status text: Title Medium below button
- Stats card: bg-secondary, 16dp radius, 16dp margin
- Bottom section: Speed + data, Body Small

6.2.2 Server Selection

- Full screen with search AppBar
- Search field: 56dp, bg-secondary
- Quick connect: "Optimal Location" tile, 72dp
- Favorites section: Horizontal scroll chips
- Country groups: Expandable
- Items: 72dp ServerListTile
- Sort options: Latency (default), Load, Name, Distance
- Filter: Protocol, Streaming, P2P

6.2.3 Settings

- Sections: Connection, Auto-Connect, Security, App, Account, About
- Section header: Overline, Primary color
- Items: Material ListTile, 56dp height
- Toggle items: Switch trailing
- Navigation items: Chevron trailing
- Sub-screens: Push navigation (slide in from right)

6.2.4 Split Tunneling Configuration

- Tab bar: Off | Include | Exclude
- Mode explanation: Info banner at top
- App list: Checkbox tiles with app icons
- Search: Filter installed apps
- Save button: Bottom sticky, full width

- iOS limitation: Show warning about app enumeration restrictions

6.2.5 Connection Stats

- Real-time speed: Display size, Mono
- Session duration: Mono Large, counting
- Total data: Download/Upload counters
- Graph: 5-minute sparkline, full width
- Protocol details: Expandable card
- Server info: Location, IP, load

6.2.6 Support / Help

- Search FAQ at top
 - Categories: Cards in grid (2 columns)
 - Contact options: Chat, Email
 - Diagnostics: "Send logs" button
 - Version info: Caption at bottom
-

7. Interaction Patterns

7.1 Connection Flow Animation

DISCONNECTED	->	CONNECTING	->	CONNECTED
Red dot		Spinning ring		Green glow
Static		Amber color		Pulse animation
				Checkmark appears

Phase 1: Initiate (0-500ms)

- Button press: Scale to 0.95, 100ms
- Ring color transition: Red -> Amber, 200ms
- Spinner appears: Fade in + start rotation

Phase 2: Handshake (500ms-2000ms)

- Spinner: Continuous rotation (360deg/1500ms)
- Status text: "Connecting..." fade in
- Optional: Progress ring around button

Phase 3: Success (2000ms+)

- Ring color: Amber -> Green, 300ms
- Spinner fades out, checkmark scales in (0->1, 400ms bounce)
- Glow effect: 20px blur, Green at 30%, pulse loop begins
- Status: "Connected" text + server info slide up

- Stats panel: Slide in from bottom, 400ms

Disconnect Flow:

- Reverse of connect
- Green -> Red transition, 300ms
- Stats panel slides out
- Glow fades

7.2 State Transitions

Transition	Duration	Easing	Visual
Button press	100ms	ease-out	Scale 0.95
Color change	200ms	ease-in-out	Crossfade
Panel slide	400ms	cubic-bezier(0.4, 0, 0.2, 1)	translateY
Fade in	300ms	ease-out	opacity 0->1
Scale in	400ms	spring(1, 80, 10)	scale 0->1
Page push (mobile)	300ms	ease-in-out	translateX
Bottom sheet	300ms	ease-out	translateY

7.2.1 Full Connection-State Visual Treatment (Canonical 7-State Machine)

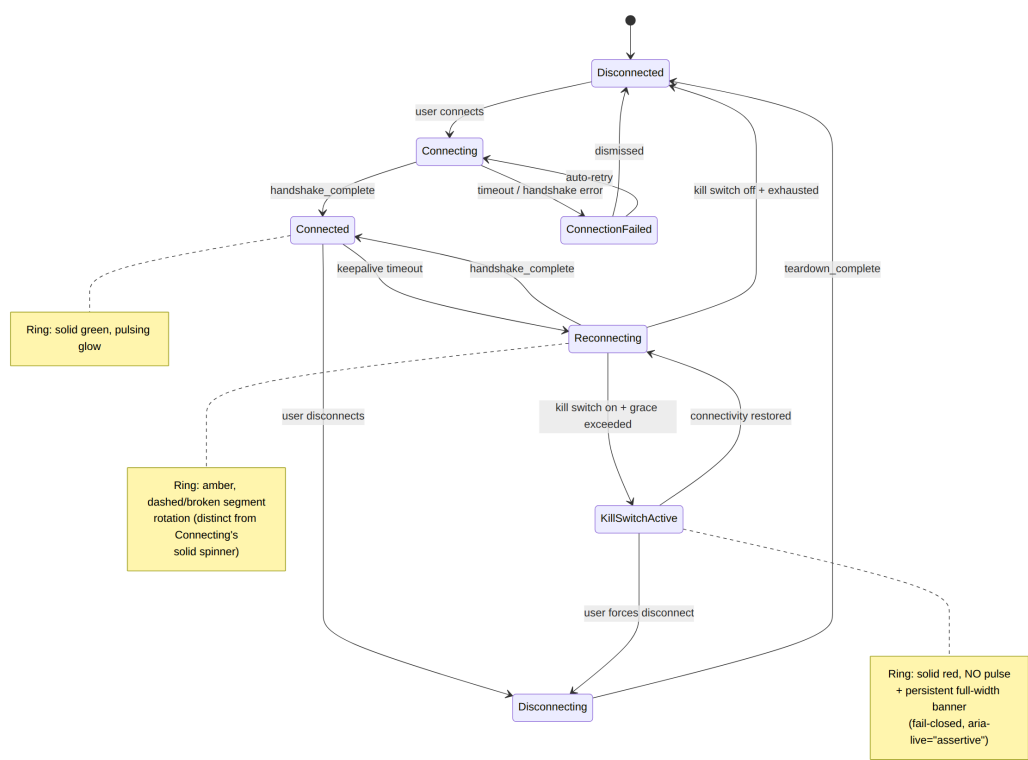
§7.1's Phase 1-3 connect animation and "Disconnect Flow" cover only 3 of the 7 states in the canonical connection lifecycle state machine owned by helix-vpn-engine (MVP2_ARCHITECTURE.md §5.6, mirrored in MVP2_SHARED_CORE.md's `ConnectionStatus` enum, Revision 2). No platform UI may invent additional states or skip a distinct treatment for any of the following — every platform's connection button, status text, glow/ring, and screen-reader announcement MUST switch over exactly this set:

State	Ring/Glow	Motion	Status Text	Screen-Reader Announcement
Disconnected	--hx-semantic-disconnected (#F44336), static, no glow	Static shield-off icon	"Disconnected"	"Disconnected. Not protected."
Connecting	--hx-semantic-connecting (#FF9800)	360°/1500ms	"Connecting..."	"Connecting to <server>..."

State	Ring/Glow	Motion	Status Text	Screen-Reader Announcements
Connected	ring, solid rotating spinner --hx-semantic-connected (#4CAF50), pulsing glow	Checkmark scale-in, pulse loop	"Connected"	"Connected to <server>, <n> milliseconds latency"
Reconnecting	--hx-semantic-connecting (#FF9800) ring, dashed/broken segment rotation — deliberately distinct from Connecting's solid spinner	Dashed ring segment rotates, no checkmark	"Reconnecting..." (never plain "Connecting...")	"Connection lost, reconnecting automatically. Your traffic is paused."
KillSwitchActive	--hx-semantic-error (#D32F2F) solid ring, no pulse (deliberately static/alarming) + full-width banner	Static lock/shield-blocked icon	"Kill Switch Active — All traffic blocked" (persistent banner, never a toast)	"Kill switch active. All network traffic is blocked until the VPN reconnects." (aria-live="assertive interrupts other announcements)
Disconnecting	Transitional --hx-semantic-connected → --hx-semantic-disconnected crossfade, 300ms	Brief spinner or simple fade, no checkmark	"Disconnecting..."	Skip announcement if state is <300ms (avoid AT chatter)
ConnectionFailed	--hx-semantic-error (#D32F2F),	Alert-triangle icon	"Connection Failed" + inline retry button	"Connection failed."

State	Ring/Glow	Motion	Status Text	Screen-Reader Announcements
	static, one-time shake micro-interaction on entry (300ms, respects reduced-motion)			<reason>. Retry available

Kill Switch banner (all platforms): a full-width, non-dismissible-until-state-exits banner using `--hx-semantic-error` at full opacity (not the 12%/18% tint used for inline hints elsewhere in this system) — the one semantic state where the strong, non-tinted color is intentional, since it communicates an active traffic block rather than an informational hint. Desktop: banner docked below the title bar. Mobile: banner docked below the app bar, persists across navigation while the state holds. Aurora: banner replaces the pulley-menu label area (§5.3.1). See MVP2_SECURITY_PERFORMANCE.md §2 for the fail-closed firewall behavior this banner reports on.



This diagram is the UI-layer rendering companion to the authoritative engine state diagram in MVP2_ARCHITECTURE.md §5.6 — transition labels are abbreviated here for visual-treatment clarity; the engine document is authoritative for exact trigger conditions.

7.3 Pull-to-Refresh

- Trigger: Pull down past 64dp on scrollable lists
- Indicator: Circular progress, Primary color
- Release threshold: 80dp
- Animation: Rotation + arc sweep
- Haptic: Light impact at threshold (iOS)
- Completion: Snap back 200ms

7.4 Swipe Actions (Mobile)

Direction	Action	Target
Swipe left (server)	Delete favorite	Server list item
Swipe right (server)	Quick connect	Server list item
Swipe left (setting)	Reset to default	Settings item

- Action width: 72dp
- Background: Semantic color (green for connect, red for delete)
- Icon: 24dp, white
- Full swipe: Auto-execute action

7.5 Long-Press Menus

- Trigger: 500ms long press
- Menu: Context menu / Bottom sheet (adaptive)
- Items: Copy IP, View details, Quick actions
- Haptic: Medium impact on trigger
- Dismiss: Tap outside or back gesture

7.6 Keyboard Navigation (Desktop)

Key	Action
Tab	Move focus forward
Shift+Tab	Move focus backward
Enter/Space	Activate focused element
Escape	Close modal/dialog/menu
Ctrl+K	Quick connect/disconnect
Ctrl+Shift+S	Open server selection
Ctrl+,	Open preferences
Ctrl+Q	Quit application

- Focus ring: 2px Primary outline, 2px offset
- Focus visible: Only on keyboard navigation (not mouse click)

7.7 Screen Reader Support

- All interactive elements have descriptive aria-label
- Connection state changes announced via aria-live="polite"
- Server selection announces: "Connected to US East, 23 milliseconds latency"
- Status updates: "Connection lost, reconnecting"
- Landmark regions: main, navigation, complementary
- Skip links for main content

7.8 Per-Platform Screen-Reader Bridge Mechanics

§7.7 states the content-level requirements (aria-label, aria-live, landmarks). Five of this system's 8 platforms are **not** native-widget UI trees (Tauri = WebView-hosted HTML/CSS/JS, Flutter = Skia/Impeller-rendered widget tree, QML = Qt Quick scene graph), so the accessibility tree the OS screen reader walks is not automatically populated the way a native UIKit/AppKit/Win32 tree is — each framework requires an explicit bridge:

Platform	UI Tree	OS Screen Reader	Bridge Mechanism
macOS / Windows / Linux (Tauri)	WKWebView / WebView2 / WebKitGTK (HTML DOM)	VoiceOver / Narrator / Orca	The DOM role/aria-* attributes from §7.7 are read directly by the WebView's built-in accessibility adapter (WKWebView exposes the DOM tree to VoiceOver via AXWebArea; WebView2 via UIA; WebKitGTK via AT-SPI) — no additional native bridge code needed, but a <div> with only a click handler and no role/aria-label is invisible to the OS tree regardless of platform.
Android / iOS / HarmonyOS (Flutter)	Skia/ Impeller-rendered widget tree (not a native view hierarchy)	TalkBack / VoiceOver / HarmonyOS screen reader	Flutter's Semantics widget tree is compiled by the engine into the platform's native accessibility tree (SemanticsNode → Android AccessibilityNodeInfo / iOS UIAccessibilityElement).

Platform	UI Tree	OS Screen Reader	Bridge Mechanism
Aurora OS (Qt6/QML)	Qt Quick scene graph	Orca (via AT-SPI, where Aurora exposes it) / Aurora's Silica accessibility layer	Every custom-painted widget (the connection toggle's glow ring, the speed-graph sparkline, §5.2.8) MUST be wrapped in an explicit <code>Semantics(label:, value:, button: true, liveRegion: true)</code> — Flutter does not infer semantics from pixels; an un-annotated <code>CustomPaint</code> is a silent accessibility hole. Each interactive QML Item MUST set the <code>Accessible</code> attached property (<code>Accessible.role</code>, <code>Accessible.name</code>, <code>Accessible.description</code>) — Qt Quick items have no accessibility representation by default; custom Item subclasses used for the connection toggle (§5.3) and cover page (§5.3.7) MUST implement <code>QAccessibleInterface</code> or set the attached properties explicitly. Standard DOM ARIA, same mechanism as Tauri's <code>WebView</code> — the popup and options page ARE plain web pages read natively by the browser's own accessibility tree.
Web Extension (React)	Browser DOM	NVDA / JAWS / VoiceOver / Orca (via the host browser)	

Verification note: a bridge that is wired but never exercised is unverifiable — screen-reader coverage MUST be confirmed by actually running the platform's screen reader against the rendered build (VoiceOver/TalkBack/Orca/NVDA), never by code-reading the `Semantics/aria-*/Accessible` annotations alone.

7.9 Dynamic Type / Font-Scaling Support

All ten typography tokens (§3.2) and four monospace tokens (§3.4) MUST scale with the platform's user-controlled text-size setting — a fixed-px implementation that ignores OS-level scaling is a WCAG 2.1 AA 1.4.4 (Resize Text) failure:

Platform	Mechanism	Scale Range	Layout Behavior at Max Scale
macOS / Windows / Linux (Tauri)	CSS rem units (already used throughout §3.2) respond to the OS/browser base font-size	Up to 200% (WCAG 1.4.4 minimum)	Server list item (§5.1.3) and settings row (§5.1.6) grow to accommodate wrapped labels rather than truncating; the 140px connection button does not scale — it is icon-only, text lives outside it
Android (Flutter)	MediaQuery.textScaler — every Text widget MUST read the ambient text scaler, never a hardcoded fontSize	Up to 200% (Android "Font size" + "Display size", compounding)	Bottom nav labels truncate to icon-only beyond 150% scale; server list switches to two-line subtitle layout
iOS (Flutter/Cupertino)	MediaQuery.textScaler mapped from UIContentSizeCategory, including the accessibility sizes (AX1-AX5, up to ~310%)	Up to ~310% (Dynamic Type accessibility range)	At AX sizes, CupertinoListTile rows support multi-line mode; the 140dp connection button remains fixed-size (icon+glow only)
HarmonyOS (Flutter+ArkUI)	MediaQuery.textScaler, HarmonyOS system font-scale setting	Up to 200%	Same reflow rules as Android
Aurora OS (QML/Silica)	Silica Theme.fontSizeXXXXXX tokens already scale with Aurora's system-wide "Font size" setting; QML text	Aurora's 5-step system font-size setting	Silica list items (§5.3.3) grow height automatically

Platform	Mechanism	Scale Range	Layout Behavior at Max Scale
	elements MUST bind to <code>Theme.fontSize*</code> , never a hardcoded px		per Silica's own layout rules

7.10 Minimum Touch Target Sizes (Mobile)

Per §1.4, all interactive elements on touch platforms (Android/iOS/HarmonyOS/Aurora) MUST provide a minimum hit target of **44×44dp/pt** (WCAG 2.1 AA 2.5.5), independent of the element's *visual* size:

- Server list favorite star (visually 20px, §8.2) → hit target padded to 44×44dp via invisible tap-area expansion, not a visually larger icon.
- Protocol badge (visually 20px tall, §5.1.4/§5.2.x) is informational-only, never itself a tap target — the entire server list row (64dp/72dp, already ≥44dp) is the tap target.
- Swipe-action buttons (§7.4, 72dp width) already exceed the floor.
- Connection toggle (140dp) exceeds the floor by a wide margin.
- Toast dismiss [X] (§5.1.9, visually 16px icon) → 44×44dp tap area required despite the toast's compact 48px height — pad the touch zone around the icon rather than shrinking the toast to fit a small icon.

7.11 Accessibility Testing Checklist (Per Phase)

Extends Appendix B with a dedicated per-phase accessibility gate — a phase does not exit "done" until its row here is checked (see Appendix B for the phase-by-phase checklist this table pairs with):

Phase	Accessibility Gate
Phase 1 (Foundation)	Color contrast validated ≥4.5:1 body / ≥3:1 large-text + UI components, both themes, all custom palettes (§1.2 OpenDesign presets)
Phase 2 (Core Components)	Every component has a keyboard-operable equivalent (§7.6) + an <code>aria-label</code> / <code>Semantics/Accessible.name</code> equivalent (§7.8)
Phase 3 (Screen Layouts)	Landmark regions present (desktop/web); Flutter <code>Semantics</code> traversal order matches visual reading order; QML <code>Accessible</code> focus order matches visual order
Phase 4 (Interactions)	Reduced-motion fallback verified per animation (§9.5); dynamic-type reflow verified

Phase	Accessibility Gate
Phase 5 (Polish)	at 150%/200%/max-scale (§7.9) on every screen
	Live screen-reader pass — VoiceOver (macOS+iOS), TalkBack (Android), NVDA + Orca (Linux), Narrator (Windows) — against the actual rendered build, not code inspection alone; 44×44dp touch-target audit (§7.10) on every tappable element

8. Iconography

8.1 Icon Set Requirements

Unified icon set across all platforms:

- Style: Outlined (line weight 1.5px-2px), rounded caps
- Size variants: 16px, 20px, 24px, 32px, 48px
- Color: Inherits text color by default
- Platform icon libraries:
 - Desktop/Web: Lucide React (or Phosphor Icons)
 - Mobile (Android): Material Symbols Outlined
 - Mobile (iOS): SF Symbols
 - Aurora: Sailfish Silica icons

8.2 Core Icon Inventory

Icon	Name	Usage	Platforms
Power	power / power_settings_new	Connection toggle	All
Shield	shield / shield.checkered	Security status	All
Shield Check	shield-check	Protected state	All
Globe	globe	Server/ location	All
Server	server	Server list	All
Settings	settings / gearshape	Preferences	All
Chevron Right	chevron-right	Navigation	All
Chevron Down	chevron-down	Expand	All
Search	search / magnifyingglass	Search	All

Icon	Name	Usage	Platforms
Close	x / xmark	Close/dismiss	All
Check	check / checkmark	Selected	All
Star	star / star.fill	Favorite	All
Clock	clock	Duration/time	All
Speed	gauge / speedometer	Speed test	All
Download	download / arrow.down	Download data	All
Upload	upload / arrow.up	Upload data	All
Lock	lock	Secure/encrypted	All
Lock Open	lock-open	Unsecured	All
Wifi	wifi	Network	All
Wifi Off	wifi-off	No network	All
Alert	alert-triangle	Warning	All
Info	info	Information	All
Trash	trash	Delete	All
Refresh	refresh-cw	Refresh/retry	All
Menu	menu	Hamburger menu	All
Help	help-circle	Support	All
Log Out	log-out	Sign out	All

8.3 Platform-Specific Icons

System Tray Icon (Desktop)

State	macOS	Windows	Linux
Disconnected	Template PNG (B&W)	.ico 16x16	PNG 22x22
Connected	Template + green dot	.ico + overlay	PNG + badge
Connecting	Template + amber dot	.ico + overlay	PNG + badge
Error	Template + red dot	.ico + overlay	PNG + badge

- macOS: Use template image for dark mode compatibility
- Badge dot: 8px diameter, bottom-right offset
- Tooltip: "HelixVPN - Connected (US East)"

App Icon (All Platforms)

- Primary: Teal (#00897B) shield/globe motif
- Format: SVG source, exported to all platform sizes
- Sizes: 16, 32, 48, 72, 96, 128, 144, 192, 256, 512, 1024px
- Shape: Adaptive (Android), Rounded corners (iOS), Square (desktop)
- Background: Transparent or Primary-900

Notification Icon

- Android: Monochrome vector drawable, white on notification background
- iOS: App icon with tinted overlay
- Desktop: Platform-native notification with app icon

Quick Settings Icon (Android)

- Size: 24dp (standard QS icon)
- Style: Monochrome, white fill
- States: Disconnected (outline), Connected (filled)
- Label: "Helix VPN"

Extension Icon (Browser)

- Toolbar: 16px, 32px (PNG)
 - Extension page: 48px, 128px
 - Manifest: 16, 32, 48, 128px
 - Color: Follows connection state badge overlay
-

9. Animation & Motion

9.1 Connection Animation

Button Pulse (Connected State):

Animation: scale 1.0 -> 1.08 -> 1.0
Duration: 1500ms
Easing: ease-in-out
Iteration: Infinite
Target: Outer glow ring + shadow

Ripple Effect (On Connect):

Origin: Button center
Animation: Scale from 0 to 4x, opacity 0.5 -> 0
Duration: 600ms

Easing: ease-out
Color: Primary at 20%

Status Glow:

Connected: Green glow, 20px blur, pulse opacity 0.2 -> 0.4
Connecting: Amber glow, rotating sweep gradient
Disconnected: No glow, static red
Error: Red pulse, 800ms (faster than connected)

9.2 Page Transitions

Desktop:

- Modal: Fade in backdrop (200ms) + scale content (300ms, spring)
- Page switch: Crossfade 200ms
- Drawer: Slide from right, 300ms ease-out

Mobile (Android):

- Push: Slide in from right (enter) + fade out (exit), 300ms
- Pop: Slide out to right (exit) + fade in (enter), 300ms
- Bottom sheet: Slide up 300ms, drag dismiss with velocity

Mobile (iOS):

- Push: Slide in from right, 350ms, Cupertino transition
- Pop: Slide out to right, 350ms
- Modal: Cover vertical, 400ms

9.3 Loading States

Skeleton Loading:

- Background: Surface color
- Shimmer: Linear gradient sweep, 1200ms
- Shape: Rounded rectangles matching content
- Trigger: Data fetch > 300ms

Spinner:

- Size: 24px (small), 40px (medium), 56px (large)
- Stroke: 3px, Primary color
- Duration: 1000ms per rotation
- Track: Surface variant, 20% opacity

Progress Bar:

- Height: 4px (linear), 48px (circular)
- Fill: Primary gradient
- Track: Surface variant

- Animation: Smooth width transition, 300ms

9.4 Micro-Interactions

Interaction	Trigger	Animation	Duration
Button hover	Mouse over	Scale 1.02, subtle lift	150ms
Button press	Mouse down	Scale 0.98	100ms
Toggle switch	Tap	Thumb translateX + color	200ms
Checkbox	Tap	Checkmark stroke draw	200ms
Radio	Tap	Dot scale in (spring)	200ms
Card hover	Mouse over	translateY -2px, shadow increase	200ms
List item tap	Touch	Ripple from touch point	400ms
Refresh	Pull	Rotation + arc sweep	1000ms
Toast	Trigger	Slide in from top-right	300ms
Copy feedback	Tap copy	Checkmark flash	500ms

9.5 Reduced Motion Support

All animations respect `prefers-reduced-motion: reduce`:

```
@media (prefers-reduced-motion: reduce) {  
  *, *::before, *::after {  
    animation-duration: 0.01ms !important;  
    animation-iteration-count: 1 !important;  
    transition-duration: 0.01ms !important;  
  }  
}
```

Reduced Motion Fallbacks:

- Pulse -> Static glow (no animation)
 - Page transitions -> Instant
 - Spinners -> Static indicator + text "Loading..."
 - Ripples -> Instant opacity change
 - Graphs -> Static (no draw animation)
-

10. Platform Adaptations

10.1 macOS

Aesthetic Principles:

- Safari-style minimal chrome
- Translucent sidebar support (vibrancy)
- Rounded window corners (system default)
- System font: SF Pro

Native Integration:

- MenuBarExtra: Icon + status tooltip, click to show window
- Native menu bar: HelixVPN > About, Preferences, Services, Quit
- Window: Title bar with traffic lights, no custom chrome
- Keyboard shortcuts: Cmd+K (connect), Cmd+, (prefs), Cmd+Q (quit)
- Notifications: NSUserNotification with action buttons
- Auto-start: Login item via SMLoginItemSetEnabled

Design Adjustments:

- Increased border radius (16px cards vs 12px default)
- Subtle shadows (macOS design language)
- Sidebar navigation for settings (3-column layout optional)
- Translucent toolbar: `window.vibrancy = 'under-window'`

10.2 Windows

Aesthetic Principles:

- Fluent Design influence (Mica/Acrylic materials where supported)
- Segoe UI font family
- Corner radius: 8px (Windows 11) / 0px (Windows 10)
- System context menu styling

Native Integration:

- System tray: NotifyIcon with context menu
- Jump list: Connect, Disconnect, Recent servers
- Notifications: Windows Toast with inline actions
- Installer: MSI with WiX, UAC elevation
- Auto-start: Registry Run key

Design Adjustments:

- Snap layouts: Window supports Windows 11 snap zones
- Title bar: Custom with Mica material on Win11

- Settings: NavigationView-style left sidebar
- Density: Slightly more compact (8px spacing vs 12px)

10.3 Linux

Aesthetic Principles:

- Adaptable to GTK/Qt themes
- Respects system accent color where possible
- Follows freedesktop.org HIG
- Font: System default (usually Cantarell, Noto Sans)

Native Integration:

- System tray: AppIndicator3 / StatusNotifierItem
- Desktop file: Categories=Network;Security;VPN;
- Notifications: dbus notification server
- Auto-start: .desktop file in ~/.config/autostart
- Theming: Detect GTK theme, adapt colors

Design Adjustments:

- Respect GTK_THEME for color adaptation
- Flat design (no shadows) for GNOME compatibility
- Header bar: Integrate with CSD if available
- Density: Configurable (compact/comfortable)

10.4 Android (Material You)

Aesthetic Principles:

- Material 3 design system
- Dynamic color: Primary derived from wallpaper (Android 12+)
- Rounded shapes: 12dp cards, 24dp dialogs
- Elevation: Surface tint instead of shadows

Native Integration:

- Quick Settings tile: 1x1 toggle in notification shade
- Foreground service: Persistent notification with controls
- App shortcuts: Long-press launcher icon for connect/disconnect
- Widget: 4x1 home screen widget with status + toggle
- Biometric: Fingerprint/face unlock for app access
- Always-on VPN: SUPPORTS_ALWAYS_ON metadata

Design Adjustments:

- Dynamic color: Use dynamicColorScheme() when available
- Fallback: Static teal palette on Android <12
- Navigation: Bottom nav bar (3 items: Connect, Servers, Settings)

- Status bar: Transparent with dark/light icon adaptation
- Edge-to-edge: Extend content behind system bars

10.5 iOS (Cupertino Design)

Aesthetic Principles:

- Cupertino design language
- SF Symbols iconography
- Blur effects: System materials (systemThinMaterial, etc.)
- Rounded corners: System default radii
- Font: SF Pro Text/Display

Native Integration:

- Control Center: VPN toggle in Settings > VPN
- Widget: Small + medium home screen widgets
- Shortcuts: Siri Shortcuts for "Connect VPN"
- Biometric: Face ID / Touch ID gate
- Push: APNs for service alerts
- URL scheme: helixvpn://connect, helixvpn://disconnect

Design Adjustments:

- Navigation: CupertinoNavigationBar with blur
- Lists: CupertinoFormSection.insetGrouped for settings
- Buttons: CupertinoButton with system styling
- Switches: CupertinoSwitch (slimmer than Material)
- Pickers: CupertinoPicker / modal bottom picker
- Modals: CupertinoModalPopup with sheet presentation

10.6 HarmonyOS (ArkUI Design Language)

Aesthetic Principles:

- ArkUI design language
- Smooth rounded corners: 16vp cards, 24vp dialogs
- Subtle gradients and depth
- Font: HarmonyOS Sans

Native Integration:

- Service widget: Form card for home screen quick connect
- VpnExtensionAbility: System VPN integration
- Notification: HMS Push Kit for alerts
- Distributed capability: Cross-device sync (future)
- Biometric: System fingerprint/face

Design Adjustments:

- Colors: Teal primary with HarmonyOS accent adaptation
- Cards: Elevated with subtle shadow

- Typography: HarmonyOS Sans throughout
- Animations: Smooth spring physics (ArkUI default)
- Dialogs: Centered with blur backdrop

10.7 Aurora OS (Silica)

Aesthetic Principles:

- Silica UI design language
- Dark-first design (ambiance integration)
- Edge gestures: Swipe from edges for navigation
- Pulley menus: Pull down for context actions
- Transparency and blur effects
- Font: Sailfish Silica

Native Integration:

- Cover page: 1x1 app cover with connection status
- Pulley menu: Pull down for connect/disconnect/server
- Ambiance: Adapt UI tint to system ambiance
- Sailfish notifications: Platform-native alerts
- D-Bus: System integration for network state

Design Adjustments:

- Transparency: Silica glass effect for panels
 - Navigation: No bottom nav, use page stack + gestures
 - Input: Silica-styled text fields with underline focus
 - Lists: Sailfish list items with press overlay
 - Primary action: Often in pulley menu, not always visible
 - Page transitions: Silica horizontal slide with depth
-

11. Localization & Internationalization (i18n) UX

11.1 RTL Layout Mirroring

Helix VPN ships RTL locale support (Arabic, Hebrew, Persian, Urdu) across all 8 platforms. RTL is a full layout mirror, not a text-direction-only change — the following components have an explicit RTL variant:

Component	LTR Behavior	RTL Behavior
Navigation chevron (\$8.2 chevron-right)	Points right (forward = deeper into hierarchy)	Points left — swap to a dedicated chevron-left icon asset, do not CSS-transform-mirror a right-pointing glyph (produces a

Component	LTR Behavior	RTL Behavior
Server list row (§5.1.3/§5.2.2)	Flag+name left, chevron+badges right	visually "wrong" arrowhead on some icon sets) Full row mirror: flag+name right, chevron+badges left — <code>dir="rtl"</code> (Tauri/React) / <code>Directionality.rtl</code> (Flutter) / <code>LayoutMirroring.enabled</code> (QML)
Connection-flow animation (§7.1/§7.2.1)	N/A	Unchanged — the circular connect/disconnect ring is direction-agnostic by construction; explicitly exempted from RTL variants
Swipe actions (§7.4, mobile)	Swipe left = delete, swipe right = quick-connect	Swapped: swipe right = delete, swipe left = quick-connect (swipe-to-reveal direction follows reading direction, not a fixed screen-edge convention)
Push navigation transitions (§9.2)	Push slides in from right	Push slides in from left
Speed graph / sparkline (§5.2.8)	Time flows left→right (oldest→newest)	Time flows right→left (oldest→newest); axis labels re-anchor
Data usage indicator (§5.2.9)	Download left, Upload right	Download right, Upload left
Toast slide-in (§5.1.9)	Slides in from top-right	Slides in from top-left

Numeric/monospace content (IP addresses, latency in ms, protocol badges "WG"/"OV"/"IK", key fragments) is **never mirrored** — digits and Latin-script protocol abbreviations retain LTR internal reading order even inside an RTL row, per the Unicode Bidi Algorithm's numeric-run handling. Wrap these spans in `dir="ltr"` (Tauri/React) or an explicit `TextDirection.ltr` override (Flutter) so the bidi algorithm doesn't reorder digit groups.

11.2 String Length Variance

English strings are the design baseline, but German and Finnish equivalents commonly run 30–50% longer (e.g. "Connection Failed" → "Verbindung fehlgeschlagen"; "Kill Switch Active" →

"Tappokytikin aktiivinen"). The layout grid (§4) accommodates this — no string-bearing component may hard-clip text with overflow: hidden / TextOverflow.clip as its only behavior:

- **Buttons (§5.1.1):** width is content-driven (min-width floor only), not fixed-width; a translated label exceeding $\sim 2\times$ the English baseline wraps to two lines (min-height: 40px becomes a floor, not a ceiling) rather than truncating or overflowing.
- **Server list item title (§5.1.3/§5.2.2):** single-line ellipsis truncation is acceptable **only** for the city/region name; the state label ("Connected"/"Reconnecting"/etc., §7.2.1) is **never** ellipsis-truncated — a truncated safety-critical state word is a trust-defeating failure mode for a security product.
- **Settings rows (§5.1.6/§6.2.3):** the 56px row height is a floor, not a ceiling — rows with a wrapped two-line label grow to 72px, divider grows with it.
- **Toast notifications (§5.1.9):** auto-width up to 400px already; translations exceeding that wrap to a second line and grow height from 48px rather than truncating; auto-dismiss timing (4000ms) is held constant regardless of line count.
- **Kill Switch banner (§7.2.1):** the highest-stakes string in the system — explicitly exempted from any truncation; no max-height, wraps freely.

11.3 Locale-Aware Number & Date Formatting (Statistics Screens)

The Connection Stats / Connection Details screens (§6.1.5, §6.2.5) MUST format per the active locale, not per a hardcoded en-US convention:

- **Decimal/thousands separators:** "1.2 GB" (en-US) vs "1,2 GB" (de-DE, fr-FR) — resolved via the platform's locale-aware number formatter (Intl.NumberFormat for Tauri/React, NumberFormat from intl/flutter_localizations for Flutter, Qt.locale().toString() for QML) — never a hand-rolled string template with a hardcoded . or ,.
 - **Date/duration display:** session duration (Mono Large, "02:34:18") is locale-invariant and stays as-is; any *calendar date* shown (e.g. subscription renewal date in Settings > Account) MUST use the locale's date format (DD.MM.YYYY vs MM/DD/YYYY vs ISO 8601) via the platform's date formatter, never a fixed pattern.
 - **RTL numeral note:** locales expecting Eastern Arabic numerals may render them via the platform's locale formatter; the monospace stat displays (§3.4) fall back to the platform's default numeral rendering for these glyphs, since the monospace technical font stack does not guarantee Eastern-Arabic-numeral coverage — verified per-locale during Phase 5 (§7.11) rather than assumed.
-

12. Biometric Authentication Fallback UX

Android, iOS, and HarmonyOS gate app access behind biometric auth (§10.4/§10.5/§10.6). This section specifies the on-screen flow those platform sections reference but do not detail: failure, unavailability, non-enrollment, and lockout.

12.1 First-Time Explainer Screen ("Why Is This Needed")

Shown once, before the first biometric prompt is ever issued (not on every app launch):

```
+-----+
| [Shield+Fingerprint |
|   illustration]    |
|                     |
|   Protect your VPN app   |
|                     |
| Face/Touch ID keeps your |
| connection settings and |
| account private if this |
| device is shared or lost. |
|                     |
| [ Enable Face ID   ]    |
| [ Not now          ]    |
+-----+
```

- Title: Title Large. Body: Body Medium, secondary color.
- Primary button triggers the OS biometric prompt; "Not now" routes to the PIN/passcode fallback (§12.3) as the standing app-lock mechanism instead — never leaves the app entirely unlocked.
- Shown exactly once; the decision is persisted, never re-asked on cold start.

12.2 Biometric Prompt States

State	Trigger	On-Screen Treatment
Prompting	App foreground after lock timeout	OS-native biometric sheet (Android BiometricPrompt, iOS LAContext Face/Touch ID sheet) — Helix VPN does not

State	Trigger	On-Screen Treatment
		draw a custom biometric UI; a branded "waiting" screen (logo + "Unlock to continue") shows behind the OS sheet
Success	Match confirmed	OS sheet dismisses; app content fades in, 200ms
Failed (repeated)	No match / face not recognized	OS sheet shows its own native retry affordance; after 2 consecutive OS-reported failures, an inline hint appears: "Having trouble? Use PIN instead" → routes to §12.3
Unavailable (hardware absent/ disabled)	canEvaluatePolicy / BiometricManager reports no hardware or disabled at OS level	Skip the biometric sheet entirely, route straight to §12.3 — never show a prompt the OS has already reported as unusable
Not enrolled	BIOMETRIC_ERROR_NONE_ENROLLED / LAError.biometryNotEnrolled	One-time inline banner: "No Face ID/fingerprint set up on this device — using PIN" then route to §12.3; link to the OS enrollment settings page, never attempt to enroll biometrics from within Helix VPN

12.3 PIN / Passcode Fallback Screen

```
+-----+
|      Enter your PIN      |
|      [ ][ ][ ][ ][ ]   |
|      1 2 3              |
|      4 5 6              |
|      7 8 9              |
|      0 <-               |
|      Forgot PIN?        |
+-----+
```

- 4-6 digit numeric keypad; digit dots fill left-to-right on entry.
- Incorrect PIN triggers a horizontal shake micro-interaction (§9.4-style, 300ms, respects prefers-reduced-motion) + dots clear + red flash on the dot row (--hx-semantic-error).
- "Forgot PIN?" routes to account recovery — never to a biometric-bypass shortcut.

12.4 Lockout Messaging After N Failed Attempts

Failed Attempts	Treatment
1-2	Shake + clear only (§12.3), no additional messaging
3	Inline warning banner above the keypad: "2 attempts remaining before temporary lockout" (--hx-semantic-warning)
5	Keypad disables; full-screen lockout state replaces the PIN screen: "Too many attempts. Try again in 00:30." with a live countdown (Mono Medium); the timer is anchored server/OS-side, not a client-resettable timer
Countdown elapsed	Keypad re-enables automatically, warning clears, attempt counter resets
Repeated lockout cycles	Escalates to full account-level re-authentication rather than an indefinitely-repeating short lockout

Lockout copy uses --hx-semantic-warning, never --hx-semantic-error (reserved for Kill Switch, §7.2.1) — lockout signals "temporary friction," not "critical failure."

13. Multi-Account / Profile Switching UX

Helix VPN supports switching between multiple accounts on one app instance (e.g. personal + work-issued accounts) without signing out and back in.

13.1 Entry Point

- **Desktop (§6.1.4 Settings):** avatar chip in the top-right of the Settings tab bar area, always visible once ≥ 1 account is signed in — tapping opens a dropdown/popover.
- **Mobile (§6.2.3 Settings):** avatar at the top of the Settings screen's "Account" section, a tappable row labeled with the active account's display name + a "Switch" chevron.
- **Aurora (§5.3.1 Pulley Menu):** "Switch account" pull-down item, since Aurora has no persistent settings-tab-bar chrome to anchor a corner avatar.

13.2 Account-Switcher Panel

```
+-----+
| Switch account [X] |
+-----+
| (*) [Avatar] Alex - Personal |
|   Connected - US East      |
|                             |
| ( ) [Avatar] Alex - Acme Corp |
|   Not connected           |
|                             |
| [+] Add another account    |
+-----+
```

- Active account: filled/selected row (`--hx-overlay-selected`), plus its own live connection-state dot (§7.2.1 state colors) inline — the panel doubles as an at-a-glance "which account is currently tunneling" view.
- Inactive accounts show their last-known state as a muted/desaturated dot, never the live color palette.
- "Add another account" routes to sign-in (email/password, social, or §14 SSO) as an *additional* login, never a replace-current-session action.

13.3 Switch Confirmation / Loading State

1. Tap target account row → row shows an inline spinner (24px, §9.3) replacing its radio dot, label changes to "Switching..."
2. If the currently active account has a live tunnel, the switch first runs the standard Disconnecting flow (§7.2.1) for it

before loading the target profile — a switch never silently leaves an orphaned tunnel running for the account being switched away from.

3. On completion: full navigation reset to the target profile's home screen (§6.1.1/§6.2.1) — never a partial state merge between profiles' favorites, history, or settings.
4. Failure: revert to the previous account's panel with an inline error on the failed target row ("Couldn't switch — try again") — never leave the UI in an ambiguous "which account is this" state.

13.4 Per-Profile State Visual Distinction

- A persistent, small account-name/avatar chip appears in the connection screen's app bar whenever ≥ 2 accounts are signed in on the device — hidden entirely in the single-account case.
 - Server favorites (§8.2 star), connection history, and split-tunneling config (§6.2.4) are strictly namespaced per account; switching swaps the entire favorites set and history — surfaced here because its *absence* is a privacy-relevant UX defect, not merely cosmetic.
 - Each account MAY use a distinct, deterministic (hash-of-account-id) avatar tint from the OpenDesign custom-palette preset set (§1.2 — Ocean Blue/Forest/Ruby/Amethyst/Midnight) purely as a passive disambiguator in the switcher panel and app-bar chip; this is cosmetic only and **MUST NOT** replace the primary brand teal used for the actual connection-state UI, which stays teal regardless of which account is active.
-

14. Enterprise SSO Login UX

Organization-managed Helix accounts (provisioned via the MVP1 Admin API / MDM enrollment, MVP2_ARCHITECTURE.md §5.5/§10.2) authenticate via the organization's Identity Provider rather than Helix's own email/password or social login — MVP1's Authentication Service already supports OAuth2/OIDC, the protocol this flow drives.

14.1 Entry Point — "Sign In With Your Organization"

```
+-----+
|      Sign in to Helix VPN      |
|                                |
| [ Email                        ] |
| [ Password                     ] |
| [      Sign In                 ] |
```

```

|                                     |
|  -- or --                         |
|  [ G Continue with Google ]      |
|  [ A Continue with Apple   ]      |
|                                     |
|  -----                         |
|  [ Sign in with your          |
|    organization              ]      |
|                                     |
+-----+

```

- The organization SSO button is visually separated (its own divider, not grouped with social buttons) since it is a categorically different flow (domain-routed OIDC discovery, not a fixed-provider OAuth button).
- Tapping it first prompts for the user's work email/organization identifier (to resolve which IdP to route to via OIDC discovery / registered domain), then performs the handoff below — never a generic "SSO" button with no routing step.

14.2 System-Browser Handoff (Security-Driven UX Constraint)

This is a deliberate security constraint, not a missed polish opportunity: the SSO flow **MUST** open the organization's IdP login page in the platform's system browser (ASWebAuthenticationSession on iOS, Chrome Custom Tabs on Android, the OS default browser process on desktop, Qt.openUrlExternally on Aurora) — **never an embedded WebView**. An embedded WebView under the app's own control could programmatically read the credential-entry DOM or intercept form submission; a system-browser session is isolated from the requesting app, shares the user's existing IdP session cookies (enabling silent SSO), and is the only surface that can satisfy an organization-mandated browser-based second factor (hardware-key prompts, device-trust checks).

```

+-----+
| [Spinner] |
|           |
| Continuing in your browser... |
|           |
| [ Cancel ] |
|           |
+-----+

```

- The app shows this holding screen (never a blank/frozen screen) the moment the system browser is invoked, with a visible "Cancel" returning to §14.1.
- The app is backgrounded/occluded on mobile while the system browser is foreground — expected OS behavior, not a bug.

14.3 Return-to-App State

- **Mobile:** the IdP redirects via a registered custom URL scheme / Universal Link / App Link, which the OS routes back into Helix VPN; the app resumes on the holding screen (§14.2), transitions to "Signing you in..." while exchanging the auth code for tokens, then lands on the connection home screen (§6.2.1) scoped to the SSO-provisioned account.
 - **Desktop:** the IdP redirects to a local loopback URL (standard OAuth2 native-app loopback flow); the system-browser tab shows "You can close this window and return to Helix VPN," and the desktop holding screen auto-transitions the same way.
 - **Failure / user cancels in browser:** return to §14.1 with a non-alarming inline message ("Sign-in was cancelled" — --hx-semantic-info, not --hx-semantic-error, since user-initiated cancellation is not a failure state).
 - **Enterprise policy applied post-login:** any MDM-pushed ManagedPolicy (MVP2_ARCHITECTURE.md §5.5/§10.2 apply_managed_policy) takes visual effect immediately on first landing at the connection home screen — e.g. a locked server list shows a grayed-out picker with a "Managed by your organization" inline label rather than silently applying the policy with no visual acknowledgement.
-

15. Offline & Degraded-Network UI States

Three real-world network conditions are common enough to warrant first-class, purpose-built UI states rather than a generic error toast (§5.1.9) — a toast is transient and easy to miss; these are persistent conditions that shape what a user can and cannot do right now.

15.1 "No Network" (Device Has No Connectivity At All)

```
+-----+
| [Wifi-off icon]                |
|                                |
|   No internet connection      |
|   Helix VPN needs a network   |
|   connection to connect.     |
|                                |
```

```
| [ Retry ] |
+-----+
```

- Replaces the connection screen's primary content area; the connection toggle is shown disabled/grayed rather than hidden.
- Detected via the platform's native connectivity API (NWPathMonitor-class API on iOS, ConnectivityManager on Android, navigator.onLine + online/offline events for Tauri/React, Aurora's ConnMan D-Bus state) — never inferred solely from a failed connect attempt (that conflates "no network" with "server unreachable," §15.2).
- Auto-recovers the instant the OS reports connectivity restored; "Retry" remains as a manual override for flaky detection.
- Server list, favorites, and settings remain browsable (local/cached data) — only the connection toggle and network-dependent actions are disabled.

15.2 "Server Unreachable — Using Cached List"

```
+-----+
| Showing saved servers |
| Couldn't refresh - using |
| last known list (2 hours ago) |
+-----+
| [Server list, unchanged layout] |
+-----+
```

- A persistent banner (--hx-semantic-warning, not --hx-semantic-error — stale-but-usable data is degraded, not failed) docked above the server list (§6.1.3/§6.2.2), never a toast.
- States cache age using the locale-aware relative-time formatting from §11.3.
- The list renders normally from cache — latency/load figures are visually marked stale (latency dot, §5.1.5, at 50% opacity with a "cached" tooltip) rather than showing confidently-wrong live-looking numbers.
- Pull-to-refresh (§7.3) is the primary manual retry; the banner itself is also tappable and triggers the same refresh.

15.3 "Reconnecting" Banner (Distinct From the Reconnecting Connection State)

§7.2.1's Reconnecting is the VPN tunnel's own state (amber ring, dashed spinner) when a connected session's handshake drops. §15.3 covers a **different** case: the app's own background services (server-list refresh, account/session sync) losing connectivity to Helix's backend while the VPN tunnel itself may be in any state,

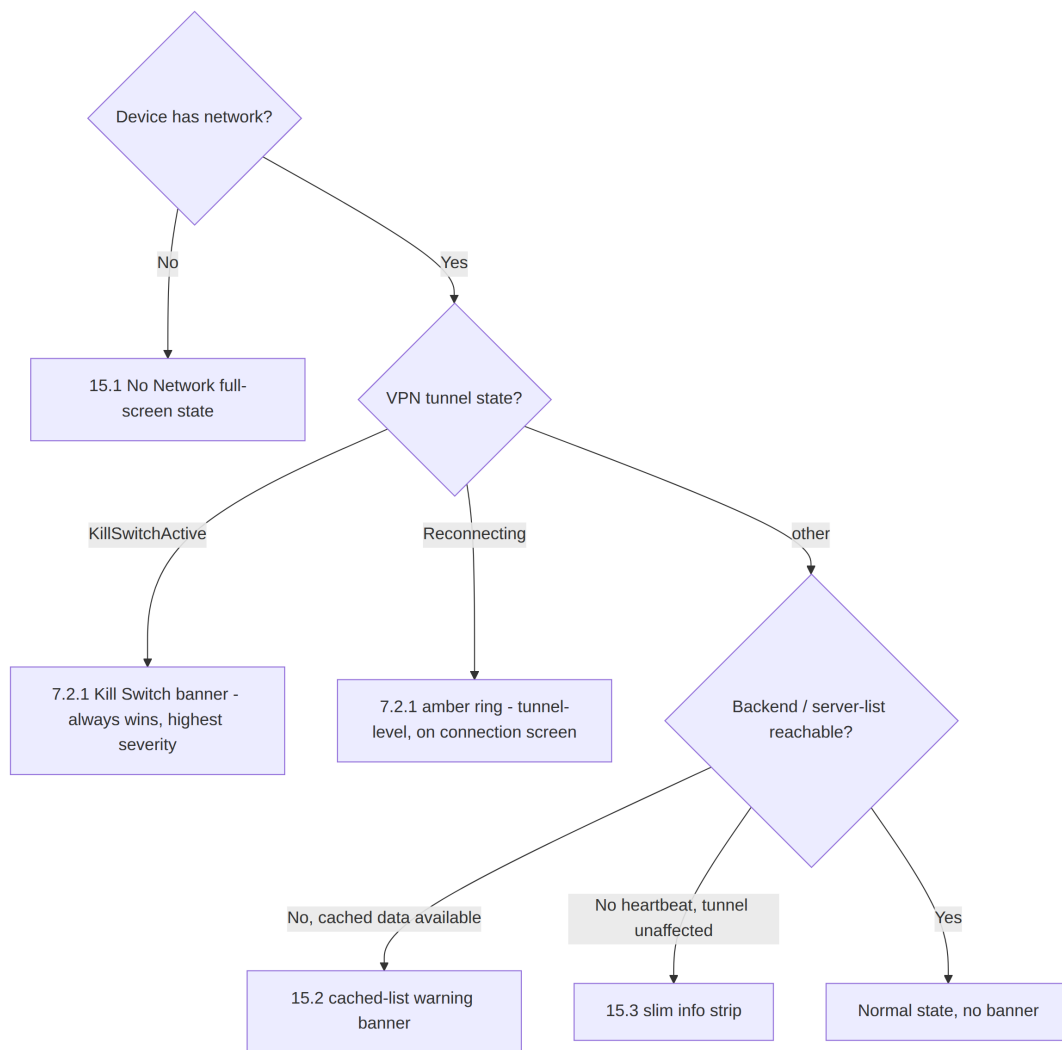
including fully Disconnected — "the app can't reach Helix's servers" as opposed to "the VPN tunnel can't reach the destination network."

```
+-----+
| [Small spinner] Reconnecting to |
| Helix services...                |
+-----+
```

- A slim, non-blocking top-of-screen strip — lighter-weight than §15.2's banner, using `--hx-semantic-info` (not warning/error), since this is an expected, usually-brief transient.
- Never appears simultaneously with the §7.2.1 Reconnecting tunnel-state ring in a way that could be misread as the same event — if both are true at once, the §15.1 "No Network" full-state treatment supersedes both.
- Auto-dismisses on successful reconnection with no persistent trace (unlike §15.2's cache-age banner, which persists until a fresh successful fetch).

15.4 State-Precedence Summary

When multiple degraded conditions could apply simultaneously, exactly one is shown — never stack multiple banners:



Appendix A: Token Reference Quick Sheet

Colors

```

/* Primary */
#00897B (500), #00796B (600), #00695C (700)
/* Accent */
#00BCD4 (400), #00ACC1 (500)
/* Semantic */
#4CAF50 (connected), #FF9800 (connecting), #F44336 (disconnected)
#D32F2F (error), #F57C00 (warning), #1976D2 (info)
/* Dark theme bg */
#0A1929 (primary), #132F4C (secondary), #1E4976 (tertiary)
/* Light theme bg */
#F0F4F8 (primary), #FFFFFF (secondary), #E8EDF2 (tertiary)

```

Spacing

4px, 8px, 12px, 16px, 20px, 24px, 32px, 40px, 48px, 64px

Border Radius

4px (sm), 8px (md), 12px (lg), 16px (xl), 24px (2xl), 9999px (full)

Animation

Fast: 100ms, Base: 200ms, Slow: 300ms, Page: 400ms

Easing: cubic-bezier(0.4, 0, 0.2, 1)

Spring: type-spring, damping: 20, stiffness: 300

Reconciled to OpenDesign token --hx-duration-fast (2026-07-04) — was 150ms, corrected to 100ms to match docs/design/opendesign/helix/tokens.css. This also matches this document's own §7.2 "Button press" (100ms) and §9.4 "Button press" (100ms) entries, which were already correct. The 150ms value survives only as the deliberately non-tokenized §9.4 "Button hover" duration, which is distinct from the named fast token and is unaffected by this fix.

Appendix B: Implementation Checklist

Each phase below carries a companion accessibility gate from §7.11 — a phase is not "done" until both its own checklist and its §7.11 accessibility-gate row are green.

Phase 1: Foundation

☐

Define all design tokens in JSON (sourced from OpenDesign to kens.css — §1.2, no independent re-derivation)

☐

Implement color system (dark + light themes)

☐

Set up typography scale per platform

☐

Configure spacing scale

☐

Implement border radius tokens

☐

§7.11 Phase 1 accessibility gate: contrast validated across both themes + all custom palettes

Phase 2: Core Components

☐

Button variants (primary, secondary, ghost, danger)

☐

Connection toggle with animation

☐

Server list item

☐

Protocol badge

☐

Latency indicator

☐

Input fields

☐

Modal/dialog

☐

Toast notification

☐

§7.11 Phase 2 accessibility gate: keyboard + screen-reader-bridge equivalent (§7.8) per component

Phase 3: Screen Layouts

☐

Desktop main window (connected/disconnected)

☐

Desktop server selection

☐

Desktop settings (tabbed)

☐

Mobile connection screen

☐

Mobile server selection

☐

Mobile settings

☐

Mobile split tunneling

☐

§7.11 Phase 3 accessibility gate: landmark regions + traversal/focus order verified

Phase 4: Interactions

- ☐ Connection flow animation (full 7-state machine, §7.2.1 — not just connect/disconnect)
- ☐ State transitions
- ☐ Pull-to-refresh
- ☐ Swipe actions (incl. RTL-mirrored direction, §11.1)
- ☐ Long-press menus
- ☐ Keyboard navigation
- ☐ Screen reader support
- ☐ §7.11 Phase 4 accessibility gate: reduced-motion + dynamic-type reflow verified at 150%/200%/max-scale

Phase 5: Polish

- ☐ Icon set implementation (incl. RTL-specific icon variants, §11.1)
- ☐ Animation system
- ☐ Reduced motion support
- ☐ Platform-specific adaptations
- ☐ Accessibility audit
- ☐ Cross-platform consistency review
- ☐ §7.11 Phase 5 accessibility gate: live screen-reader pass on rendered build (all 4 readers) + 44×44dp touch-target audit

Phase 6: Localization, Auth Edge Cases & Network Resilience (Revision 1)

- ☐ RTL layout mirror implemented + verified for every component in §11.1's table (not CSS-transform-mirrored icons)
- ☐

String-length stress test with German/Finnish pseudo-translations (§11.2) — no clipped state labels



Locale-aware number/date formatting on statistics screens (§11.3)



Biometric first-time explainer + PIN/passcode fallback + lockout messaging (§12)



Multi-account switcher entry point, panel, and per-profile state isolation (§13)



Enterprise SSO system-browser handoff (never embedded WebView) + return-to-app state (§14)



Offline / cached-list / reconnecting-banner states implemented with correct precedence (§15.4)

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Total Platforms: 8 (macOS, Windows, Linux, Android, iOS, HarmonyOS, Aurora OS, Web Extension) — corrected from "7"; Web Extension is documented throughout this file (§5.1.10, §6, §8.3) and is platform #8 in the OpenDesign coverage matrix, docs/design/README.md §2.